Zilu LIU

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https://scholar.google.com/citations?user=8HfOsw4AAAAJ&hl=en&oi=ao

Education

09/2020 – 03/2025 **The Hong Kong University of Science and Technology (HKUST)** Hong Kong S.A.R

- Ph.D., Department of Electrical and Computer Engineering
- Advisor: Prof. C. Patrick YUE (*Fellow of IEEE, Fellow of Optica*)
- GPA: 4.02/4.30.

09/2016 – 07/2020 **The South China University of Technology (SCUT)** Guangzhou, China

- Bachelor of Engineering, Information Engineering
- GPA: 3.88/4.00; Ranking: 7/243.

Professional Experience

Analog IC Design Engineer in Sanechips (ZTE) 07/2025 – Present

- Led the 448G TX front-end R&D in advanced FinFET nodes and contributed to the successful tape-out of a 224G LR SerDes.
- Proposed an architecture based on push-pull driver to achieve high bandwidth, large output swing and a new technique to achieve high linearity.
- Developing an AI-assisted automated optimization methodology for T-coil networks design.

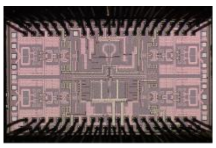
Postdoctoral Researcher in Sanechips (ZTE) 06/2025 – Present

- Led architectural exploration and silicon validation of SerDes IC for next-generation communication.
- Secured 10M RMB funding from the Shenzhen Key R&D Program for next-gen high-speed TX research.

Representative Research Projects

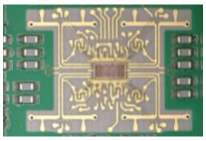
Millimeter-Wave Phase Array IC Design 07/2021 – 07/2023

- 28-GHz 4-element transmitter including IF mixer, phase shifter, PLL, VGA, RF mixer, PA.
- Tape out in TSMC-40nm process.
- Total area: 1.3 × 2.1 mm^2; P_{sat}: 20 dBm;
  Phase shifting induced gain error: < 0.04 dB; Gain control induced phase error: < 0.61°.



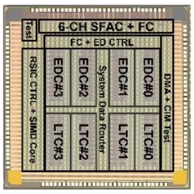
- Achieve high area and power efficiencies and orthogonal phase shifting and gain control. (Click for demo video)

Millimeter-Wave Chip-to-Antenna Matching Network Design 11/2022 – 07/2024



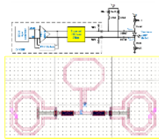
- 28-GHz flipped patch antenna with 2-stub matching network to enhance the bandwidth.
- Fabricate an 8-layer 16-element PCB.
- Increase bandwidth by 2.8 times, reaching 10.8 GHz. (Click for demo video)

Low-Power Level-Crossing ADC IC Design 03/2024 – 02/2025



- Adaptive resolution to save power of the processing.
- Tape out in TSMC-40nm LP process.
- For biomedical AI processors.

Common-Mode Filter IC Design 11/2020 – 07/2021



- 2-stage delay equalizer in DM and 2 transmission zeros in CM.
- Simulate in a 56-Gbps PAM-4 optical transmitter.
- Suppress CM noise by 18.8 dB, increase eye width of optical output by 9%.

Awards

RedBird Academic Excellence Award in 2022-23	**HKUST Fok Ying Tung Graduate School**
National Scholarship in 2016-17 and 2017-18	**Ministry of Education of the People's Republic of China**

Skills

Technical: SerDes IC design; Power amplifier design; RF board-level circuit design.

Software: Cadence, ADS, HFSS, CST, Altium Designer, MATLAB.

Language: English, Mandarin.

Other Experiences

12/2020 – Present	Co-Chair	**IEEE Solid-State Circuits Society HK Student Chapter**
08/2021 – 08/2023	Teaching Assistant Coordinator	**ECE Dept., HKUST**
12/2018 – 12/2019	Co-Chair	**Student Union, School of EE, SCUT**
04/2019 – 08/2019	EMC Analysis Engineer	**The Fifth Research Institute of MIIT, China**

Publications

** Indicates co-first authorship*

1. **Z. Liu**, L. Wang, R. Ma, Z. Chen, C. Yue, "A Compact 28-GHz Transmitter Front-End with Co-Optimized Wideband Chip-Antenna Interface," in *IEEE Transactions on Microwave Theory and Techniques*, doi: 10.1109/TMTT.2025.3561795. **(Featured Article, Top-level microwave journal)**
2. **Z. Liu***, L. Wang*, H. Fallah, Z. Chen, C. Yue, "A Compact 28-GHz Transmitter Front-End with Co-Optimized Wideband Chip-Antenna Interface Achieving 18.5-dBm P1dB and 1.0 W/mm^2 Power Density for Phased Array Systems", *2024 IEEE MTT-S International Microwave Symposium (IMS)*. **(Top-level microwave conference)**
3. L. Wang*, **Z. Liu***, R. Ma, C. Yue, "A 20-24-GHz Low-Jitter DPSSPLL with Charge Domain Bandwidth Optimization Scheme Achieving 61.3-fs RMS Jitter and -253-dB FoM_{Jitter}", *2024 IEEE Custom Integrated Circuits Conference (CICC)*. **(Top-level circuits conference)**
4. L. Wang*, **Z. Liu***, H. Fallah, R. Ma, Z. Chen, and C. P. Yue, "A 28-GHz Phased-Array Transmitter Achieving 24% Peak Efficiency, 0.26-mm^2/Element Area Efficiency with Completely Orthogonal Phase and Gain Control," in *2024 IEEE European Solid-State Electronics Research Conference (ESSERC)*, 2024, pp. 1–2. **(Top-level circuits conference)**
5. **Z. Liu**, L. Wang, H. Fallah, C. Yue, " Design of Chip-to-PCB Matching Network for Millimeter-Wave On-Chip Transmitter and On-PCB Antenna", *2023 IEEE 15th International Conference on ASIC (ASICON)*, Nanjing, China, 2023, pp. 1-4. (*Invited Paper*)
6. **Z. Liu**, R. Azmat, X. Liu, L. Wang and C. P. Yue, "On-Chip Filter for Mitigating EMI-Related Common-Mode Noise in High-Speed PAM-4 Transmitter," *2021 IEEE 14th International Conference on ASIC (ASICON)*, Kunming, China, 2021, pp. 1-4. (*Invited Paper*)
7. L. Wang, **Z. Liu**, R. Ma and C. P. Yue, "A Compact 20–24-GHz Sub-Sampling PLL With Charge-Domain Bandwidth Control Scheme," in *IEEE Journal of Solid-State Circuits*, vol. 60, no. 3, pp. 768-784, March 2025. **(Top-level circuits journal)**
8. L. Wang, **Z. Liu** and C. P. Yue, "A 24-30 GHz Cascaded QPLL Achieving 56.8-fs RMS Jitter and −248.6-dB FoM_{jitter}," *2023 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Kyoto, Japan, 2023, pp. 1-2. **(Top-level circuits conference)**
9. C. Zhang, **Z. Liu**, S. Feng, R. Ma and C. Patrick Yue, "Recent Activities of the IEEE SSCS Hong Kong Student Branch Chapter [Chapters]," in *IEEE Solid-State Circuits Magazine*, vol. 16, no. 4, pp. 125-126, Fall 2024. **(Chapter news)**
10. L. Wang, **Z. Liu** and C. P. Yue, "SSCS Hong Kong Student Chapter Holds Series of Lectures [Chapters]," in *IEEE Solid-State Circuits Magazine*, vol. 13, no. 3, pp. 50-51, Summer 2021. **(Chapter news)**
11. L. Wang, **Z. Liu** and C. P. Yue, "IEEE SSCS Hong Kong Student Chapter Activities [Chapters]," in *IEEE Solid-State Circuits Magazine*, vol. 13, no. 2, pp. 83-84, Spring 2021. **(Chapter news)**
12. F. Tian, J. Chen, K. Shao, **Z. Liu**, *et al.*, "E-NPU: A 34-126nJ/Class Event-Driven Adaptive Neural SoC with Signal-Dynamics-Aware Feature Clustering and Multi-model In-Memory Inference/Training for Personalized Medical Wearables", *2025 IEEE Custom Integrated Circuits Conference (CICC)*. **(Top-level circuits conference)**
13. C. Zhang, L. Wang, **Z. Liu**, F. Chen, Q. Pan, X. Li, C. P. Yue, "A 48-Gb/s Half-Rate PAM4 Optical Receiver with 0.27-pJ/bit TIA Efficiency, 1.28-pJ/bit RX Efficiency, and 0.06-mm^2 area in 28-nm CMOS," *2024 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Honolulu, HI, USA, 2024, pp. 1-2. **(Top-level circuits conference)**
14. W. Shi, F. Chen, X. Liu, C. Zhang, **Z. Liu**, *et al.*, "An Integrated System Evaluation Engine for Cross-Domain Simulation and Design Optimization of High-Speed 5G Millimeter-Wave Wireless SoCs," *2021 IEEE 14th International Conference on ASIC (ASICON)*, Kunming, China, 2021, pp. 1-4. (*Invited Paper*)
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